Testing pure components

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Testing pure components before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Least privilege and tool scopes

Without opening the old lesson, explain one key idea from Least privilege and tool scopes. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Configuration and secrets

Without opening the old lesson, explain one key idea from Configuration and secrets. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Requirements and service boundaries

Without opening the old lesson, explain one key idea from Requirements and service boundaries. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

> Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

Pure components return results from supplied **input**s without external side effects. They are easier to test with exact expected values.

Define remaining(capacity, used)=capacity-used for integer **input**s satisfying 0<=used<=capacity. Invalid types/ranges raise an explicit validation error. Test zero and equality boundaries.

Practise with fading help

> Complete the supported Testing pure components trace and record the first step you can explain without help.

Use W2: Trace the worked case for Testing pure components. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Testing pure components explanation against the supplied facts before discussing it.

Use the worked case for Testing pure components. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Capacity 9 with used 0, 9, 10 and Boolean True. Give outcomes.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Testing pure components facts and keep your

> reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Trace the worked case

> Trace the worked case. Apply the supplied Testing pure components facts and keep your reasoning.

Capacity 12, used 5; capacity 12, used 12; capacity 12, used 13. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Audit the limit

> Audit the limit. Apply the supplied Testing pure components facts and keep your reasoning.

A report omits this qualification: Passing unit cases for a **pure function** does not establish integration permissions or model-dependent behaviour.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Testing pure components facts and keep your reasoning.

Capacity 9 with used 0, 9, 10 and Boolean True. Give outcomes.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Testing pure components facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its **expected outcome** before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Testing pure components to Grounded Learning Service, identify a limit and choose the next useful test.

Add a labelled testing pure components **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of testing pure components.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

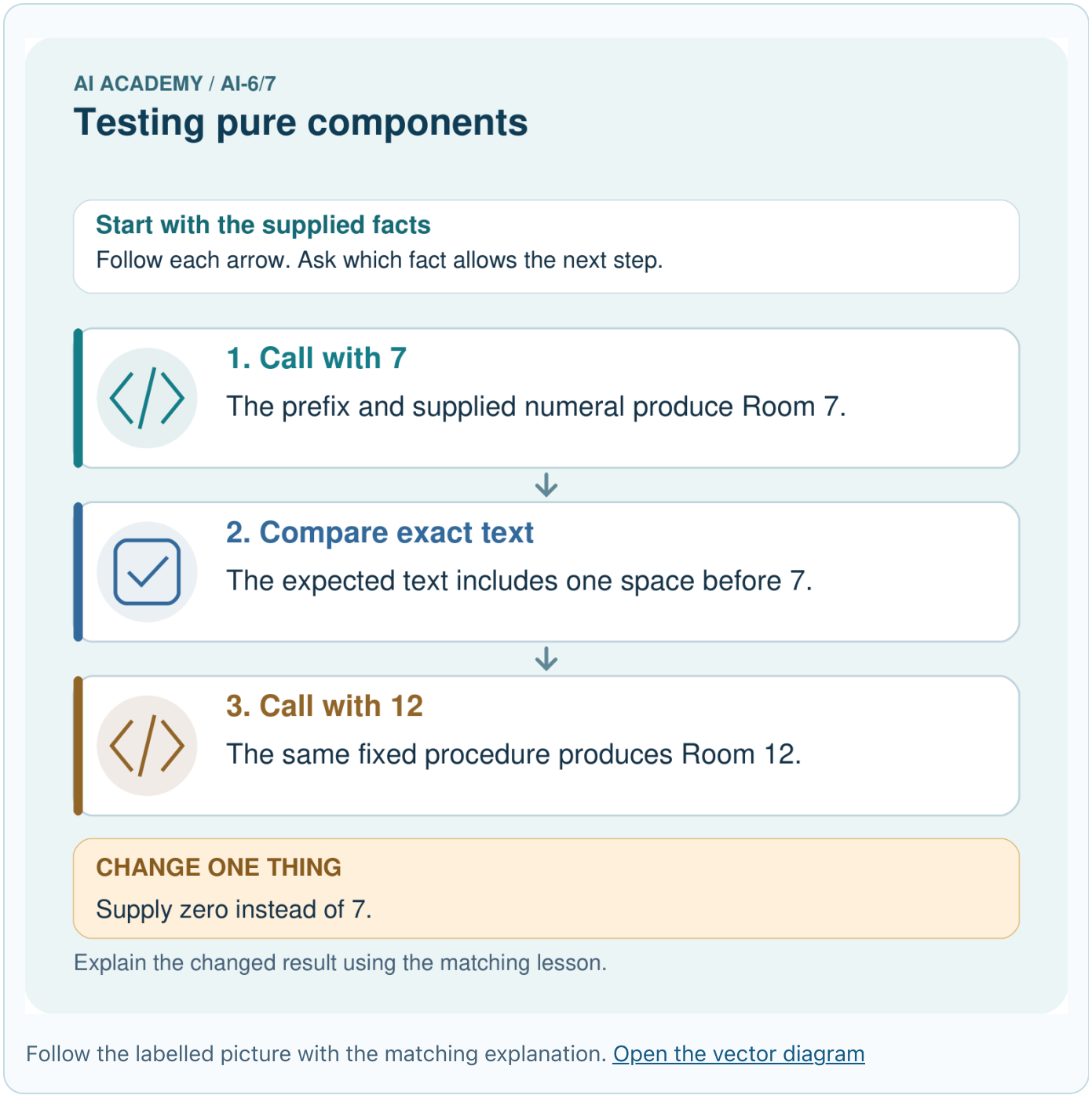
Pictorial case practice: Testing pure components

> Test a room formatter alone. A pure function returns "Room " plus a supplied positive integer.

> Explain the labelled step before checking the changed condition.

Test a room formatter alone. A **pure function** returns "Room " plus a supplied positive integer.

Accepted **input**s are positive integers. Expected **output**s: 7→"Room 7"; 12→"Room 12".



Follow the labelled picture with the matching explanation. Open the vector diagram

Predict the formatter **output**s for 7 and 12, including spacing. Why does **input** 0 need rejection under this contract rather than an ordinary formatted response?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

WORKBOOK · 13 OF 15

Project portfolio milestone

Keep this week's input, deciding step, result and limit together in the Grounded Learning Service record.

Project: Grounded Learning Service

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Test a room formatter alone. A **pure function** returns "Room " plus a supplied positive integer.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The same fixed procedure produces Room 12.

1. Call with 12

2. Compare exact text

3. Call with 7

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Call with 12 The same fixed procedure produces Room 12. A pure component depends on its accepted **input**, not a clock, network or model.

Isolating it makes failures easier to locate.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Supply zero instead of 7. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. Reject the **input** under the stated positive-integer contract.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Reject the **input** under the

> stated positive-integer contract. Zero has the right type but violates the value requirement.

Repair a missing building block

Review Least privilege and tool scopes (ai-6/6). Goal: Explain the **mechanism** and terminology of least privilege and tool scopes. Short practice: Classify read S2, write S2 and read S5 as permitted or denied.

Identify the operation or station mismatch for each denial, then reconsider after read S5 is added to trusted scope.. Return to this week and explain what you changed.

Review Configuration and secrets (ai-6/4). Goal: Explain the **mechanism** and terminology of configuration and secrets. Short practice: Point to the public setting and the protected credential.

What changes when maximum_words becomes 60, and which information must never appear in the startup log?. Return to this week and explain what you changed.

Review Requirements and service boundaries (ai-6/1). Goal: Explain the **mechanism** and terminology of requirements and service boundaries. Short practice: Split the museum visitor's request into two operations.

Which can the service perform for P, and what happens to each operation when Q has no approved card?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

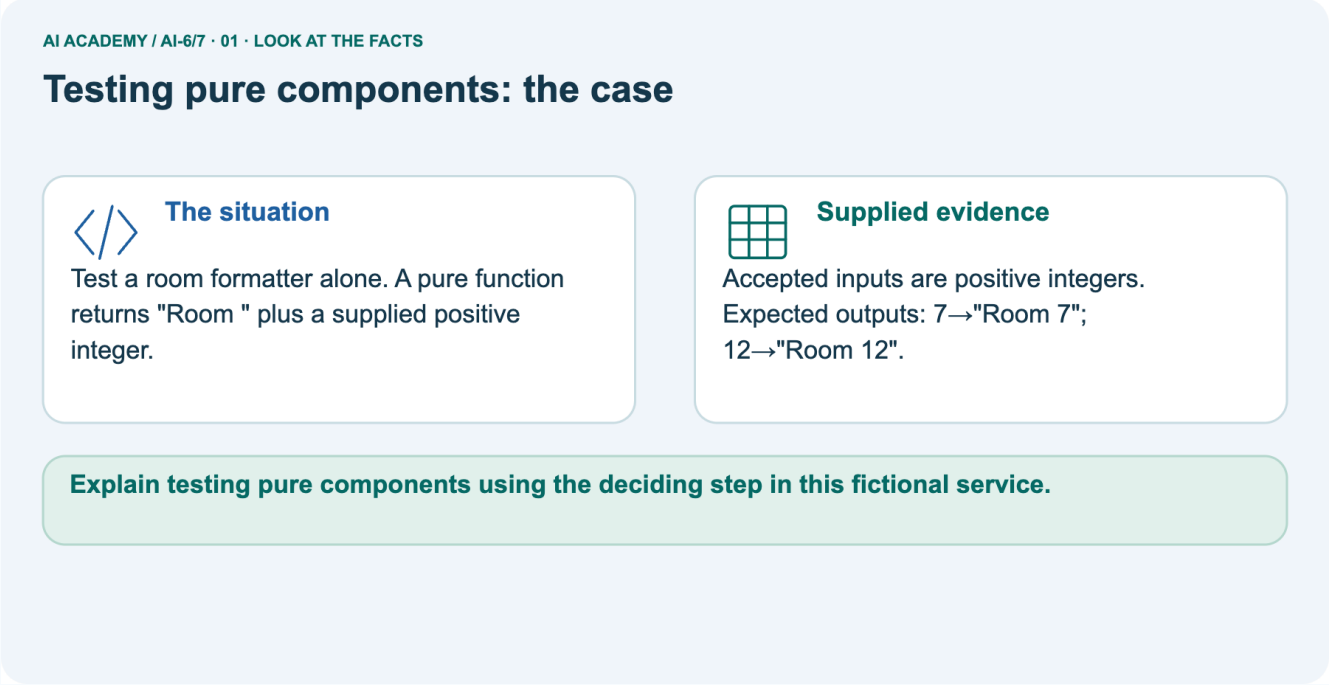
What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Testing pure components picture from its labels, then record a relevant limit.

01. Testing pure components: the case



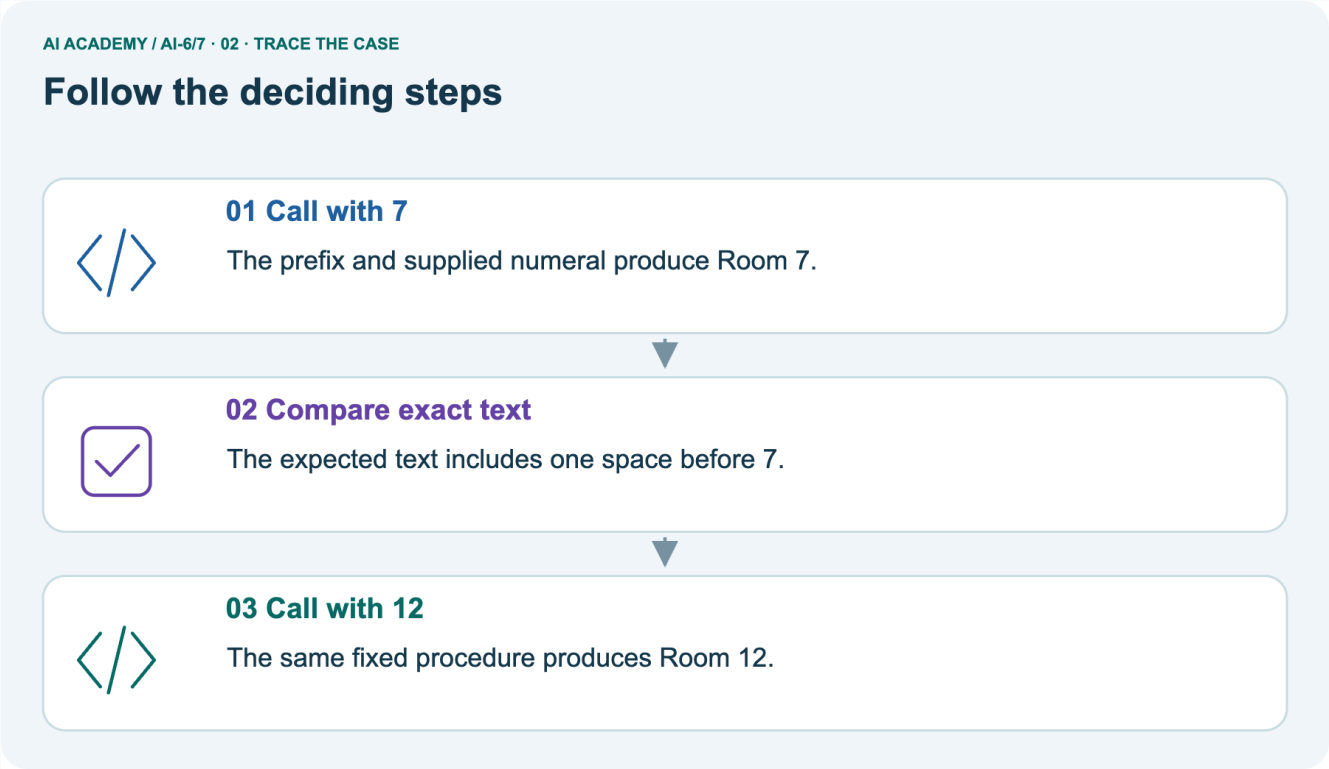
Follow the labelled picture with the matching explanation. Open the vector diagram

Test a room formatter alone. A **pure function** returns "Room " plus a supplied positive integer. Supplied facts: Accepted **input**s are positive integers. Expected **output**s: 7→"Room 7"; 12→"Room 12".

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



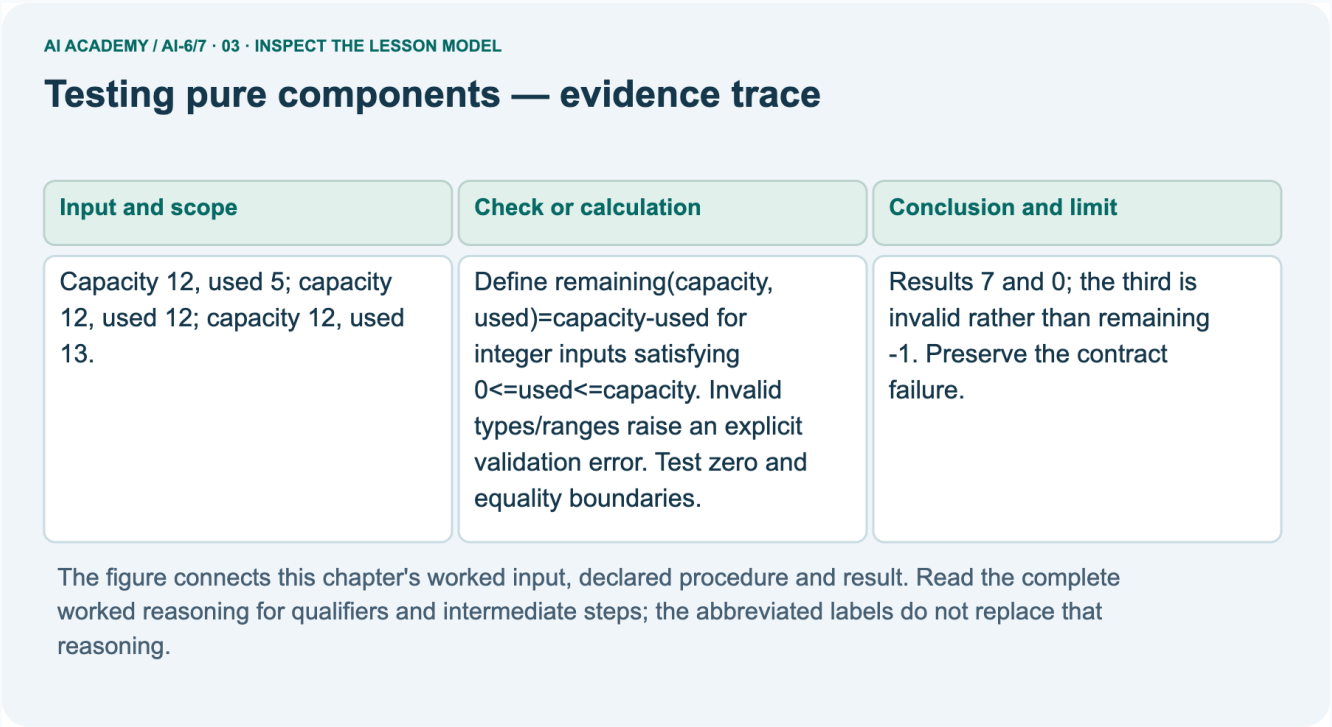
Follow the labelled picture with the matching explanation. Open the vector diagram

Call with 7: The prefix and supplied numeral produce Room 7. → Compare exact text: The expected text includes one space before 7. → Call with 12: The same fixed procedure produces Room 12.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Testing pure components — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

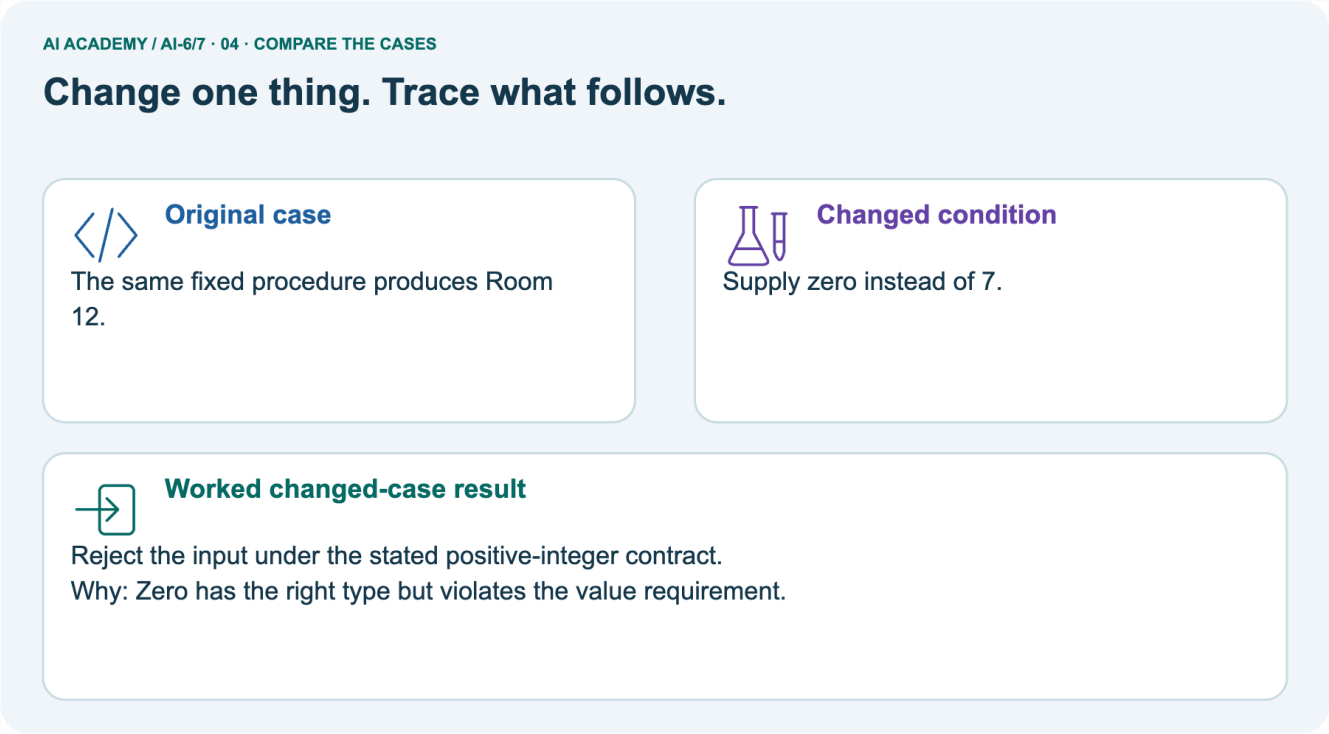
Testing pure components — **evidence** trace. Capacity 12, used 5; capacity 12, used 12; capacity 12, used 13. | Define remaining(capacity, used)=capacity-used for integer **input**s satisfying 0<=used<=capacity.

Invalid types/ranges raise an explicit validation error. Test zero and equality boundaries. | Results 7 and 0; the third is invalid rather than remaining -1. Preserve the contract failure.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

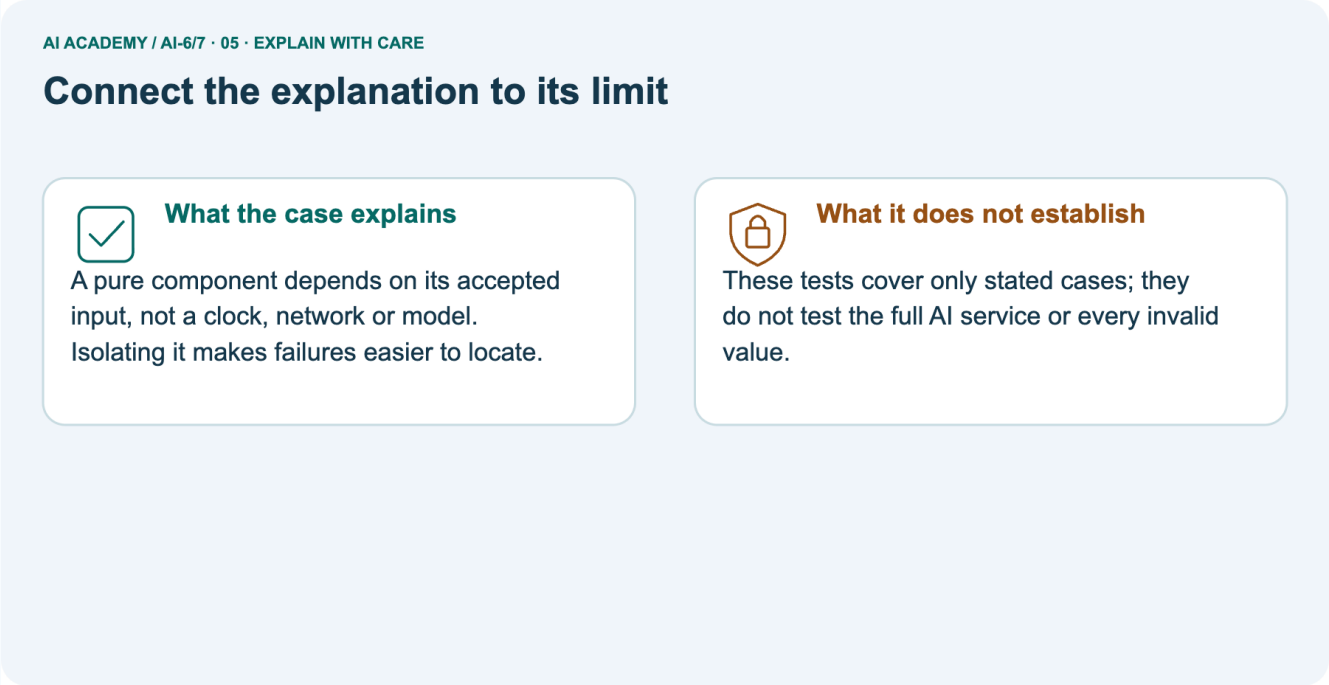
Original case: The same fixed procedure produces Room 12. Changed condition: Supply zero instead of 7. Expected result: Reject the **input** under the stated positive-integer contract.

Reason: Zero has the right type but violates the value requirement.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

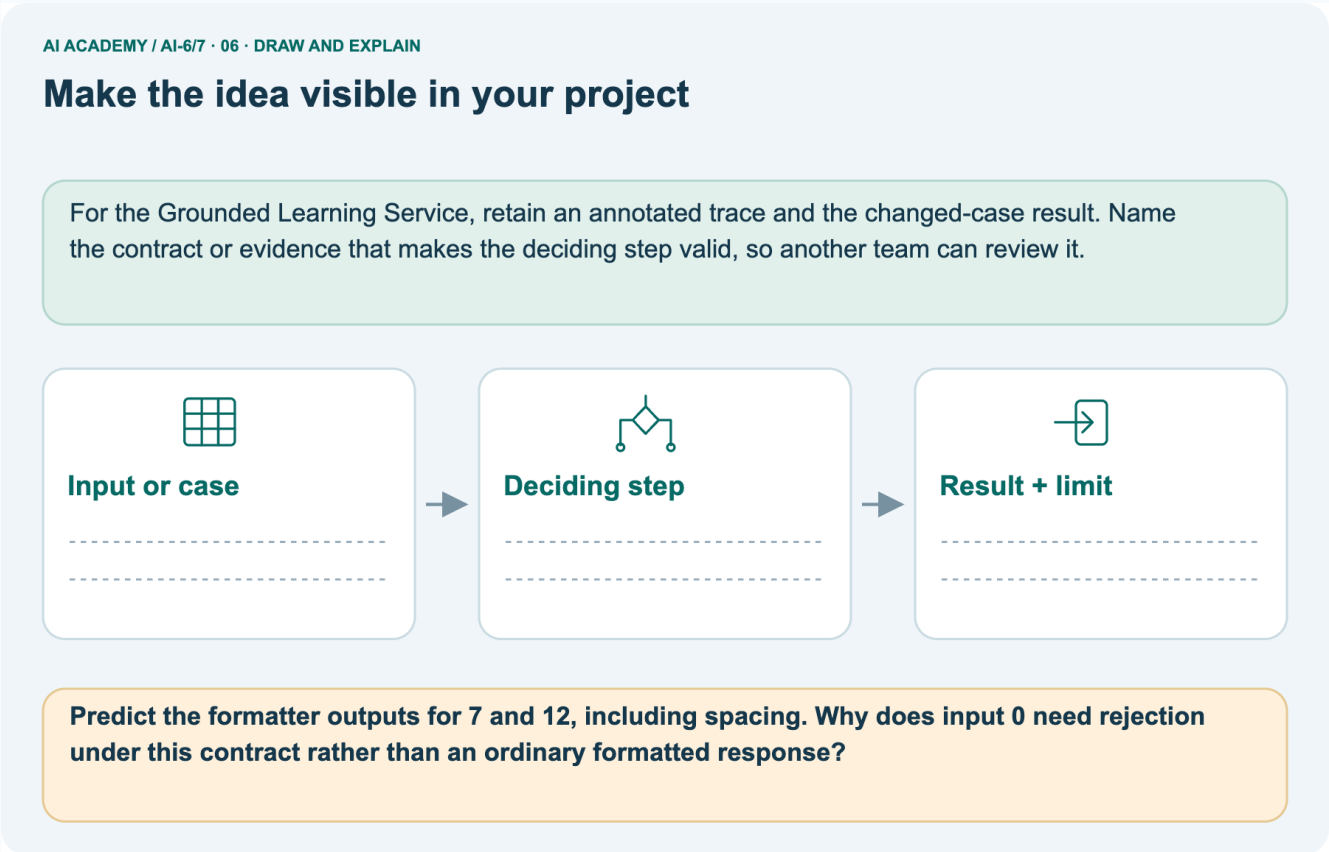
A pure component depends on its accepted **input**, not a clock, network or model. Isolating it makes failures easier to locate.

Limit: These tests cover only stated cases; they do not test the full AI service or every invalid value.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Predict the formatter **output**s for 7 and 12, including spacing.

Why does **input** 0 need rejection under this contract rather than an ordinary formatted response?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.